Supplementary Materials

S1. Extended Seed Count Robustness

The primary manuscript reports the frozen 2026-05-21 statistics. To evaluate robustness to larger seed counts, we repeated the online benchmark with 16 seeds per condition. This run pooled thalamus, distributed, and distributed meta-monitor control conditions, so it should be interpreted as a supplementary robustness check rather than as a replacement for the primary validation table.

All five constructs remained above the validation threshold (`r >= 0.70`), although the pooled estimates were lower than the primary frozen estimates for several constructs.

Construct	Primary frozen r	16-seed supplementary r	Change
Action agency	0.874	0.727	-0.147
Boundary self	0.940	0.918	-0.022
Identity-temporal self	0.850	0.749	-0.101
Action ownership	0.996	0.987	-0.009
Distributed body-schema self	0.924	0.902	-0.022

The largest change was observed for action agency. The 16-seed estimate (`r=0.727`, 95% CI `[0.662, 0.781]`) is close to the validation threshold but remains positive, statistically reliable, and above criterion. We interpret this as evidence that the primary 8-seed estimate was somewhat upward-biased, not as evidence against the construct. The action-loop mechanism effect also replicated in the extended run (`+0.698`, 95% CI `[0.695, 0.700]`).

S2. Transformer Extended Validation

We also repeated the Transformer-inspired 2x2 follow-up with 16 seeds per condition. The result reinforces the interpretation in the main manuscript: action-loop effects generalize to an attention-based substrate, while self-related validation in the Transformer implementation remains architecture-limited.

Construct	n	r [95% CI]	Interpretation
Action agency	64	0.999826 [0.999734, 0.999907]	near-ceiling convergence
Boundary self	64	0.489 [0.286, 0.674]	not validated
Identity-temporal self	64	0.610 [0.444, 0.747]	exploratory
Action ownership	64	0.999992 [0.999989, 0.999996]	near-ceiling convergence

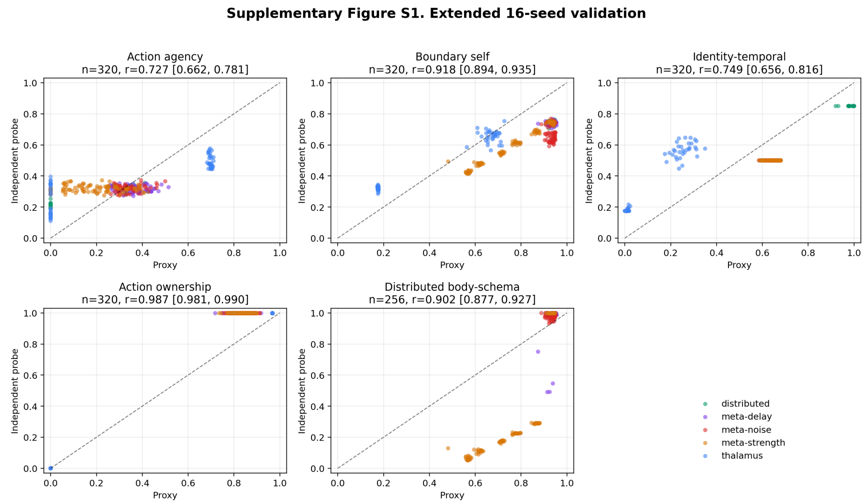

Figure 1: Supplementary Figure S1. Extended 16-seed validation.

The mechanism effects remained selective. Action-loop availability increased action agency (`+0.8190`, 95% CI `[0.8182, 0.8198]`) and action ownership (`+0.8190`, 95% CI `[0.8182, 0.8197]`). Workspace availability strongly increased the internal boundary-self proxy (`+0.8357`, 95% CI `[0.8287, 0.8426]`) but did not yield validated boundary-self convergence. Identity-temporal self also dropped into the exploratory range in this extended run. This supports a conservative conclusion: Transformer-like systems reproduce the action-loop mechanism pattern, but current self-related probes do not provide robust construct validation in this substrate.

S3. Workspace Capacity Sweeps

We ran workspace capacity sweeps in the thalamus-inspired and Transformer-inspired architectures. Each sweep tested capacities `0, 1, 2, 4, 8` with 16 seeds per capacity. The contrast between the two architectures clarifies the meaning of the Transformer measurement limitation.

S3.1 Thalamus Architecture

In the thalamus-inspired architecture, workspace capacity produced graded changes in workspace-linked measures. Proxy-independent convergence remained strong for both boundary self and identity-temporal self:

Diagnostic	n	Pearson r [95% CI]	Spearman rho
capacity vs boundary proxy	80	0.668 [0.588, 0.737]	0.826

Diagnostic	n	Pearson r [95% CI]	Spearman rho
capacity vs boundary independent	80	0.391 [0.222, 0.535]	0.343
capacity vs identity proxy	80	0.417 [0.262, 0.543]	0.271
capacity vs identity independent	80	0.508 [0.395, 0.608]	0.442
boundary proxy vs independent	80	0.838 [0.722, 0.910]	0.325
identity proxy vs independent	80	0.982 [0.965, 0.992]	0.629

S3.2 Transformer Architecture

In the Transformer-inspired architecture, workspace capacity affected the internal boundary proxy but did not rescue independent boundary validation:

Diagnostic	n	Pearson r [95% CI]	Spearman rho
capacity vs boundary proxy	80	0.494 [0.371, 0.596]	0.203
capacity vs generic boundary probe	80	-0.257 [-0.463, -0.034]	-0.078
capacity vs hard attention boundary	80	0.184 [-0.039, 0.386]	0.251
proxy vs generic boundary	80	0.404 [0.208, 0.578]	0.483
proxy vs hard attention boundary	80	0.252 [0.077, 0.435]	0.360

S3.3 Architecture-Conditioned Interpretation

The capacity sweeps show that workspace capacity is not sufficient, by itself, to establish boundary-self validity in every architecture. In the thalamus-inspired architecture, capacity-linked proxy changes corresponded to strong proxy-independent convergence. In the Transformer-inspired architecture, capacity changed internal workspace diagnostics but did not produce validated boundary-maintenance behavior. This pattern supports the main manuscript's architecture-conditioned interpretation: mechanisms may generalize while construct validity remains architecture dependent.

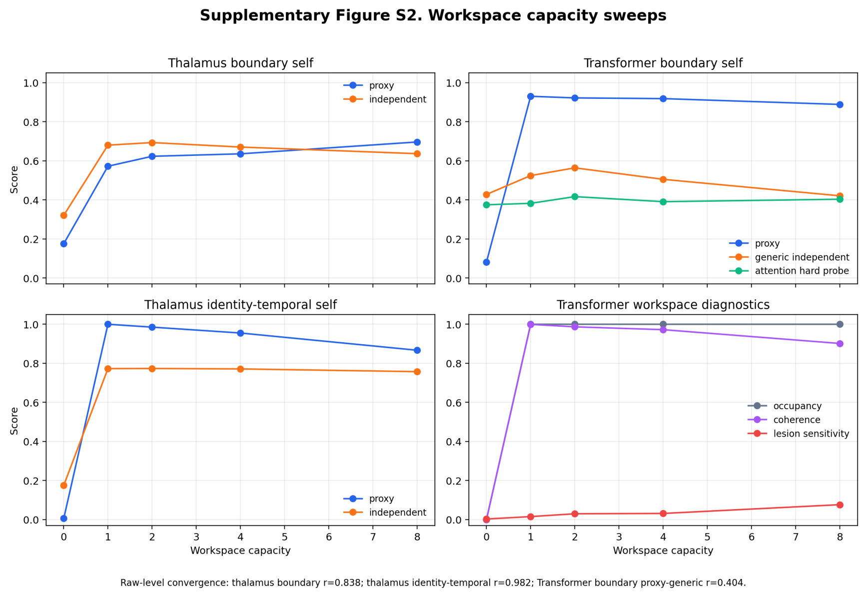

Figure 2: Supplementary Figure S2. Workspace capacity sweeps.

S4. Data Availability

Supplementary statistics and run metadata are stored in `docs/paper/statistics/supplementary_20260522/`. Supplementary figures are stored in `docs/paper/figures/supplementary_20260522/`.

The source run directories are:

- `runs/benchmark_online_supplementary/online_20260522_173251/`
- `runs/supplementary/transformer_validation_20260522_173251/`
- `runs/supplementary/transformer_workspace_capacity_20260522_173251/`
- `runs/supplementary/thalamus_workspace_capacity_20260522_173251/`

These supplementary runs are not used to overwrite the frozen primary manuscript statistics.